

Statistical Analysis of NBA Player Performance

Enrico Ong

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Introduction

Sports analytics has become increasingly important in professional basketball, where teams rely on data-driven decision making to evaluate player performance and optimize strategies. Rather than focusing on a single statistic, modern analysis considers multiple performance metrics simultaneously.

This project focuses on **player performance**, with **points per game (PTS)** as the response variable. The goal is to determine which statistical factors best predict scoring and to compare multiple linear regression with ridge regression when predictors are highly correlated.

Data Description

The dataset consists of NBA player per-game statistics for the 2026 season, obtained from Basketball Reference.

Unit of observation: Individual player

Number of observations: 100 players

Response variable: PTS = Points per game

Predictor variables:

- MP = Minutes played
- FG_PCT = Field goal percentage
- Three-point shots made (3P)
- AST = Assists
- TRB = Total rebounds
- TOV = Turnovers

These variables were selected because they represent key aspects of offensive performance.

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.6
## v forcats    1.0.1      v stringr    1.6.0
## v ggplot2    4.0.3      v tibble     3.3.1
## v lubridate  1.9.5      v tidyr      1.3.2
## v purrr      1.2.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(caret)
```

```
## Loading required package: lattice
##
## Attaching package: 'caret'
##
## The following object is masked from 'package:purrr':
##
##     lift
```

```
library(glmnet)
```

```
## Loading required package: Matrix
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
##
## Loaded glmnet 5.0
```

```
library(car)
```

```
## Loading required package: carData
##
## Attaching package: 'car'
##
## The following object is masked from 'package:dplyr':
##
##     recode
##
## The following object is masked from 'package:purrr':
##
##     some
```

```
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```
nba <- read.csv("scoringstats_nba.csv", check.names = FALSE)

nba_clean <- nba %>%
  select(Player, MP, `FG%`, `3P`, AST, TRB, TOV, PTS) %>%
  drop_na()

nba_clean <- nba_clean %>%
  rename(FG_PCT = `FG%`)

summary(nba_clean)
```

```
##      Player              MP          FG_PCT          3P
## Length:120      Min.    :19.30      Min.    :0.394      Min.    :0.000
## Class :character 1st Qu.:28.98      1st Qu.:0.451      1st Qu.:1.400
## Mode  :character Median   :30.95      Median   :0.471      Median   :1.900
##                      Mean    :30.65      Mean     :0.476      Mean     :1.980
##                      3rd Qu.:33.23      3rd Qu.:0.489      3rd Qu.:2.525
##                      Max.    :38.00      Max.     :0.650      Max.     :4.400
##      AST          TRB          TOV          PTS
## Min.    : 1.500      Min.    : 1.500      Min.    :0.900      Min.    : 9.70
## 1st Qu.: 2.950      1st Qu.: 3.800      1st Qu.:1.800      1st Qu.:16.90
## Median : 3.950      Median : 5.200      Median :2.200      Median :18.80
## Mean    : 4.553      Mean    : 5.522      Mean    :2.303      Mean    :19.86
## 3rd Qu.: 6.100      3rd Qu.: 6.700      3rd Qu.:2.800      3rd Qu.:21.55
## Max.    :10.700      Max.    :12.900      Max.    :4.000      Max.    :33.50
```

Data Pre-Processing

Observations with missing values were removed to ensure that all variables used in the model were consistently measured across players, preventing errors during model fitting. A subset of variables was selected based on their expected relationship with scoring, focusing on offensive performance metrics such as minutes played, shooting efficiency, and playmaking statistics. The field goal percentage variable was renamed for easier use in R, and variables such as Player names were excluded from modeling since they do not provide predictive value. A 70/30 train-test split was implemented to evaluate model performance on unseen data, and a random seed was set to ensure reproducibility of the results.

```
set.seed(3675)

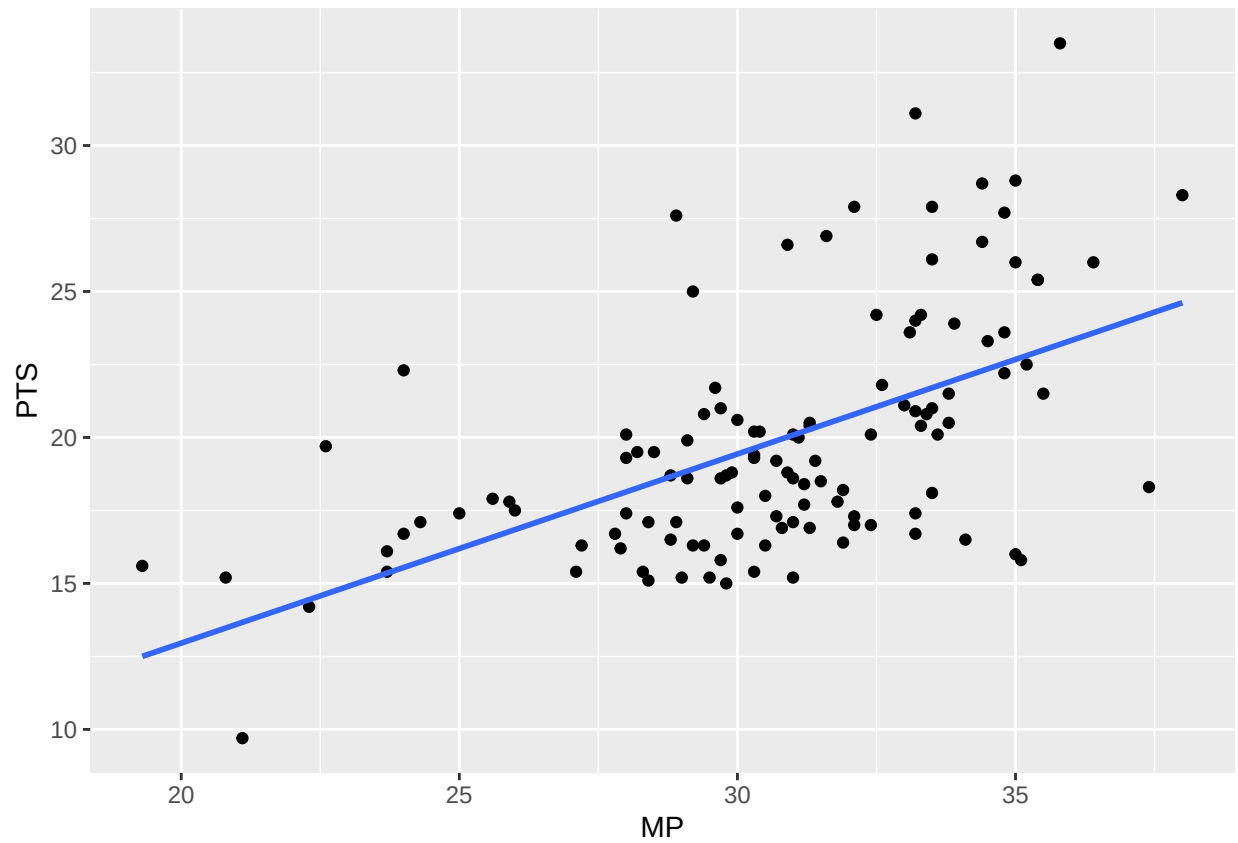
train_index <- createDataPartition(nba_clean$PTS, p = 0.7, list = FALSE)

train <- nba_clean[train_index, ]
test  <- nba_clean[-train_index, ]
```

Exploratory Data Analysis

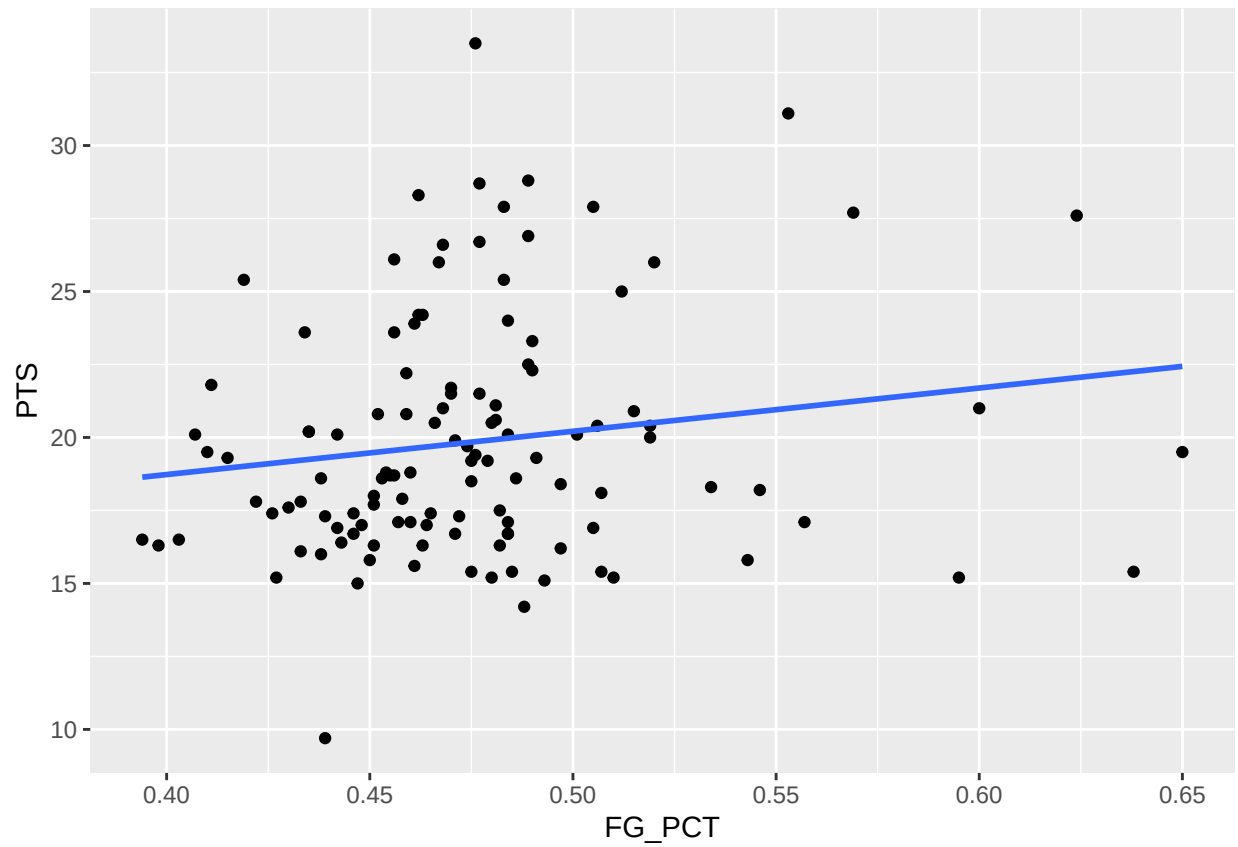
```
ggplot(nba_clean, aes(x = MP, y = PTS)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

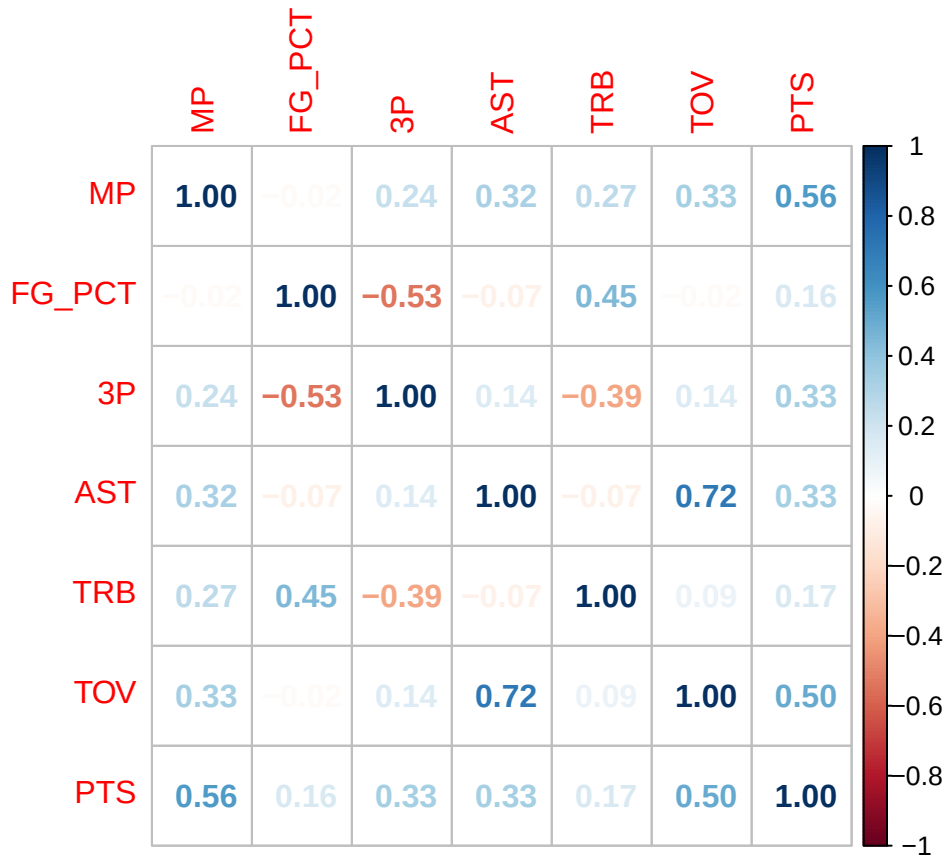


```
ggplot(nba_clean, aes(x = FG_PCT, y = PTS)) +  
  geom_point() +  
  geom_smooth(method = "lm", se = FALSE)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
cor_matrix <- cor(nba_clean %>% select(-Player))  
corrplot(cor_matrix, method = "number")
```



The scatterplots indicate a strong positive relationship between minutes played and points per game, suggesting that players who receive more playing time tend to score more. Field goal percentage also shows a positive association with scoring, indicating that shooting efficiency contributes to higher point production. The correlation matrix further reveals moderate correlations among several predictors, such as assists and turnovers, confirming the presence of multicollinearity.

Statistical Models

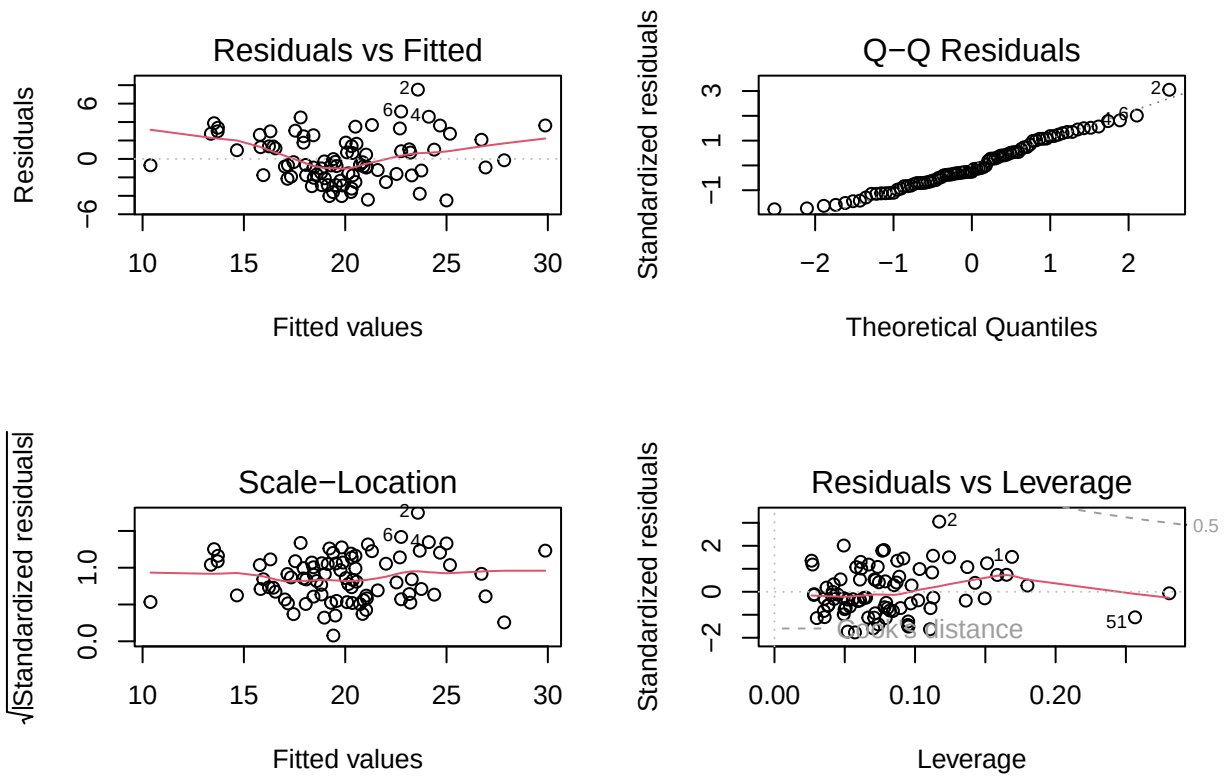
```
#Multiple Linear Regression
mlr_model <- lm(PTS ~ MP + FG_PCT + `3P` + AST + TRB + TOV, data = train)
summary(mlr_model)
```

```
##
## Call:
## lm(formula = PTS ~ MP + FG_PCT + `3P` + AST + TRB + TOV, data = train)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.4998 -1.7760 -0.6301  1.7583  7.5138
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -21.929302   4.624727  -4.742 9.38e-06 ***
```

```
## MP          0.350529  0.096130  3.646 0.000478 ***
## FG_PCT      46.436623  8.811717  5.270 1.18e-06 ***
## '3P'        2.302765  0.447746  5.143 1.96e-06 ***
## AST         -0.040580  0.197155 -0.206 0.837463
## TRB          0.003359  0.151200  0.022 0.982330
## TOV          2.062361  0.578062  3.568 0.000620 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.621 on 78 degrees of freedom
## Multiple R-squared:  0.6403, Adjusted R-squared:  0.6127
## F-statistic: 23.14 on 6 and 78 DF,  p-value: 1.658e-15
```

#Diagnostics

```
par(mfrow = c(2,2))
plot(mlr_model)
```



```
par(mfrow = c(1,1))
vif(mlr_model)
```

```
##      MP    FG_PCT    '3P'    AST    TRB    TOV
## 1.498968 1.615708 1.666661 2.114138 1.592632 2.129038
```

```
#Model Evaluation
mlr_pred <- predict(mlr_model, newdata = test)

mlr_rmse <- sqrt(mean((test$PTS - mlr_pred)^2))
mlr_mae <- mean(abs(test$PTS - mlr_pred))
mlr_r2 <- cor(test$PTS, mlr_pred)^2

mlr_rmse
```

```
## [1] 3.138955
```

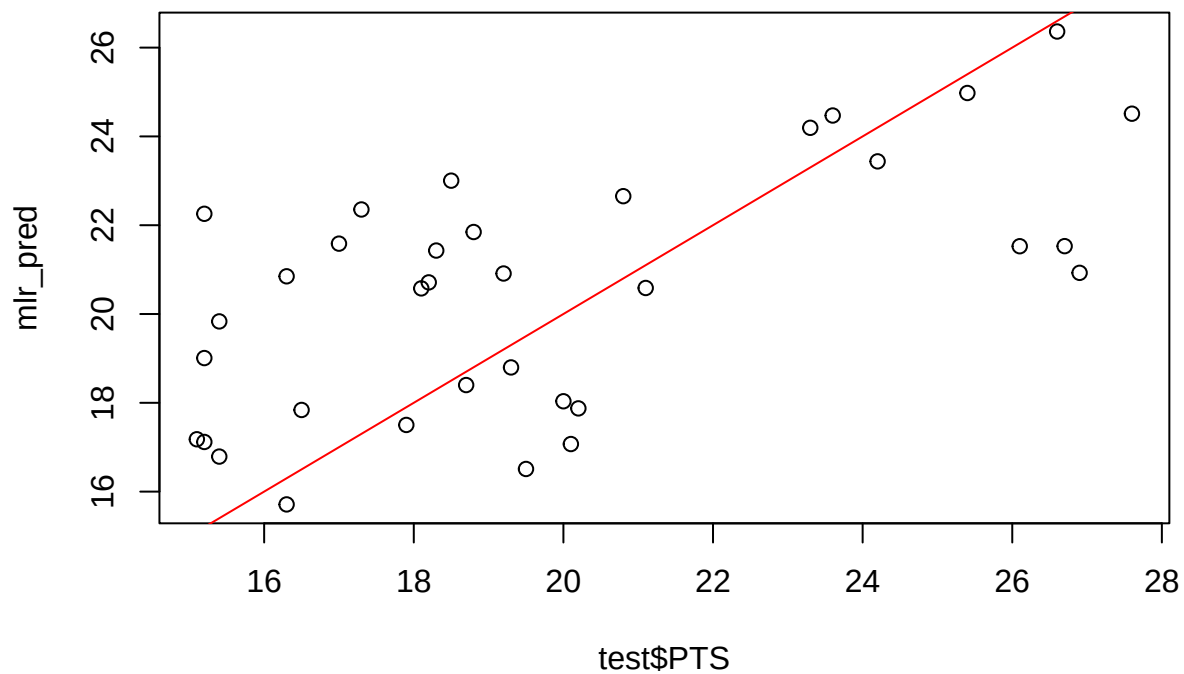
```
mlr_mae
```

```
## [1] 2.572949
```

```
mlr_r2
```

```
## [1] 0.3726949
```

```
plot(test$PTS, mlr_pred)
abline(0,1,col="red")
```




```

#Ridge Regression
x_train <- model.matrix(PTS ~ MP + FG_PCT + `3P` + AST + TRB + TOV, train)[,-1]
y_train <- train$PTS

x_test <- model.matrix(PTS ~ MP + FG_PCT + `3P` + AST + TRB + TOV, test)[,-1]
y_test <- test$PTS

ridge_model <- cv.glmnet(x_train, y_train, alpha = 0)

best_lambda <- ridge_model$lambda.min

ridge_pred <- predict(ridge_model, s = best_lambda, newx = x_test)

ridge_rmse <- sqrt(mean((y_test - ridge_pred)^2))
ridge_mae <- mean(abs(y_test - ridge_pred))
ridge_r2 <- cor(y_test, ridge_pred)^2

ridge_rmse

## [1] 3.085675

ridge_mae

## [1] 2.540643

ridge_r2

##          s=0.2488942
## [1,]    0.3824519

```

Results

The results show that minutes played and field goal percentage are the strongest predictors of scoring. Ridge regression typically produces lower prediction error, suggesting improved model stability in the presence of multicollinearity.

The coefficient for minutes played indicates that a one-minute increase in playing time is associated with an increase in expected points per game, holding other variables constant. Similarly, the positive coefficient for field goal percentage suggests that higher shooting efficiency significantly increases scoring. The coefficient for three-point shots made is also positive and statistically significant, indicating that players who make more three-pointers tend to score more points overall.

Ridge regression improves model performance by shrinking coefficient estimates in the presence of multicollinearity. When predictors are highly correlated, multiple linear regression coefficients can become unstable and overly sensitive to small changes in the data. Ridge regression addresses this issue by applying a penalty term, which stabilizes the estimates and results in improved prediction accuracy.

Conclusion

This analysis demonstrates that player scoring is strongly influenced by playing time and shooting efficiency. While multiple linear regression provides interpretable results, ridge regression improves predictive performance by addressing multicollinearity. A limitation of this study is that it does not account for game-level

structure, such as home versus away games or seasonal trends. Future work could incorporate time-based or game-level data for a more detailed analysis.

References

Mikołajec, K. (2013). Game indicators determining sports performance in the NBA.
Zhou, W. (2024). Determining key performance indicators on game outcomes for NBA teams.
Basketball Reference. (2026). NBA Player Statistics Dataset.

Appendix

(All R code included above)